TEMPEST2023_H2S

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#SET UP

```r
#Set working directory
# knitr::opts_knit$set(root.dir = "C:/Users/ashmo/OneDrive - Arizona State University/Data/TEMPEST/Pore

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(dplyr)
library(ggplot2)
library(data.table)
```

```
##
## Attaching package: 'data.table'
##
## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, mday, minute, month, quarter, second, wday, week,
##     yday, year
##
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
##
## The following object is masked from 'package:purrr':
##
##     transpose
```

```r
library(stringr)
library(plater)
```

```r
#Function to calculate slope, intercept, and R-squared and return in data frame
#Fe 2
cal_curve<-function(Conc_uM,ABS){
    model<-lm(ABS~Conc_uM)
    slope<-coef(model)[[2]]
    intercept<-coef(model)[[1]]
     R2<-summary(model)$adj.r.squared
  return(data.frame(slope=slope,intercept=intercept,R2=R2))
}
```

#IMPORT DATA

Files required: CSV for each plate run. Each plate should include a set of standards and QAQC checks (duplicates, spikes, interference)

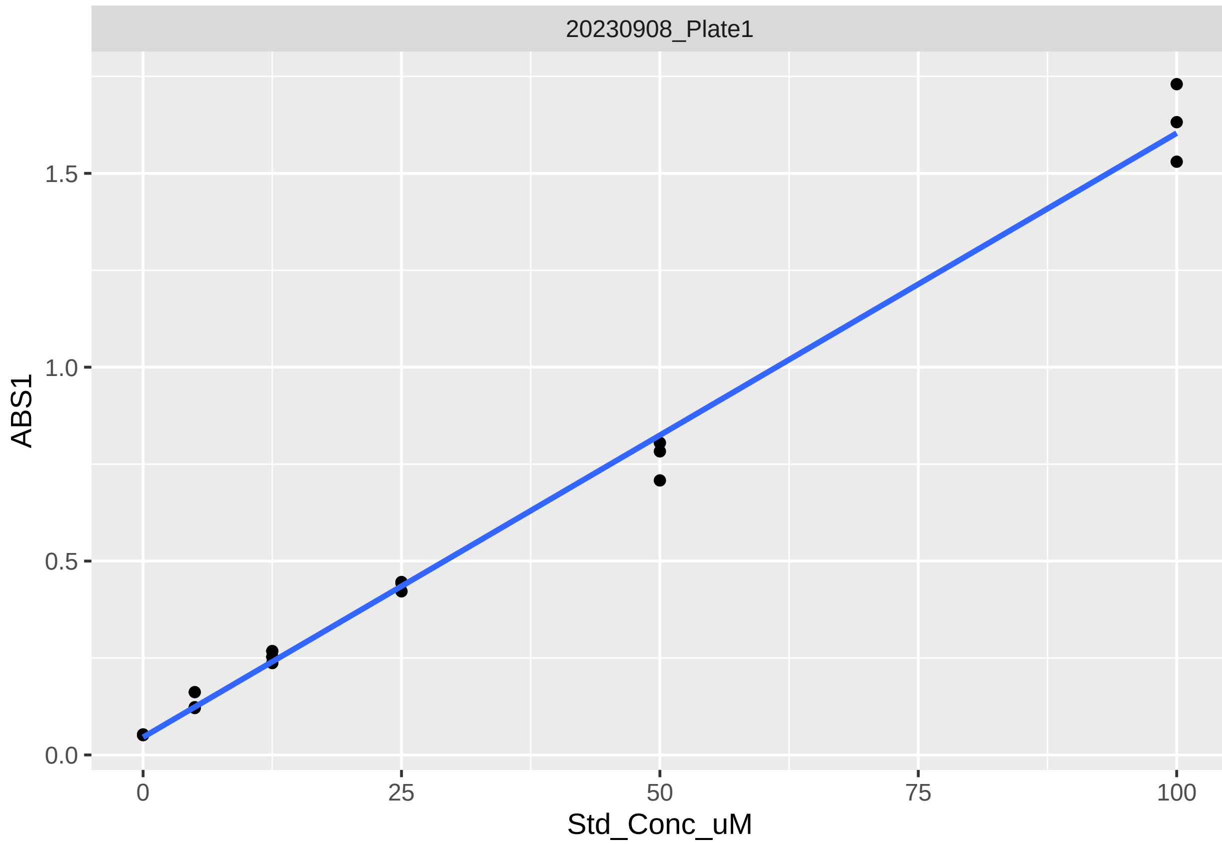
```r
# setwd("C:/Users/ashmo/OneDrive - Arizona State University/Data/TEMPEST/Porewater/02_CleanMicroplateDa

plates<-read_plates(
  files=c("Tidy Data/TEMPEST20230908_H2S_plate1.csv"),
  plate_names = c("20230908_Plate1"),
  well_ids_column = "Wells",
  sep=",")

head(plates)
```

```
## # A tibble: 6 x 6
##   Plate           Wells  ABS1 IDs            Dilutions Std_Conc_uM
##   <chr>           <chr> <dbl> <chr>              <int>       <dbl>
## 1 20230908_Plate1 A01   0.051 STD_S0                 1           0
## 2 20230908_Plate1 A02   0.053 STD_S0                 1           0
## 3 20230908_Plate1 A03   0.052 STD_S0                 1           0
## 4 20230908_Plate1 A04   0.055 20230807_SW_B4         1          NA
## 5 20230908_Plate1 A05   0.055 20230807_SW_B4         1          NA
## 6 20230908_Plate1 A06   0.06  20230807_SW_B4         1          NA
```

##PLOT STANDARDS

```r
plates%>%
  filter(str_detect(plates$IDs,"STD"))%>%
  ggplot(aes(Std_Conc_uM,ABS1))+
  facet_wrap(~Plate)+
  geom_point()+
  geom_smooth(method="lm",se=FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
stds<-plates%>%
  filter(str_detect(plates$IDs,"STD"))%>%
  group_by(Plate)%>%
  mutate(cal_curve(Std_Conc_uM,ABS1))%>%
  filter(row_number()==1)%>%
  select(Plate,slope,intercept,R2)

stds
```

```
## # A tibble: 1 x 4
## # Groups:   Plate [1]
##   Plate            slope intercept    R2
##   <chr>            <dbl>     <dbl> <dbl>
## 1 20230908_Plate1 0.0156    0.0457 0.992
```

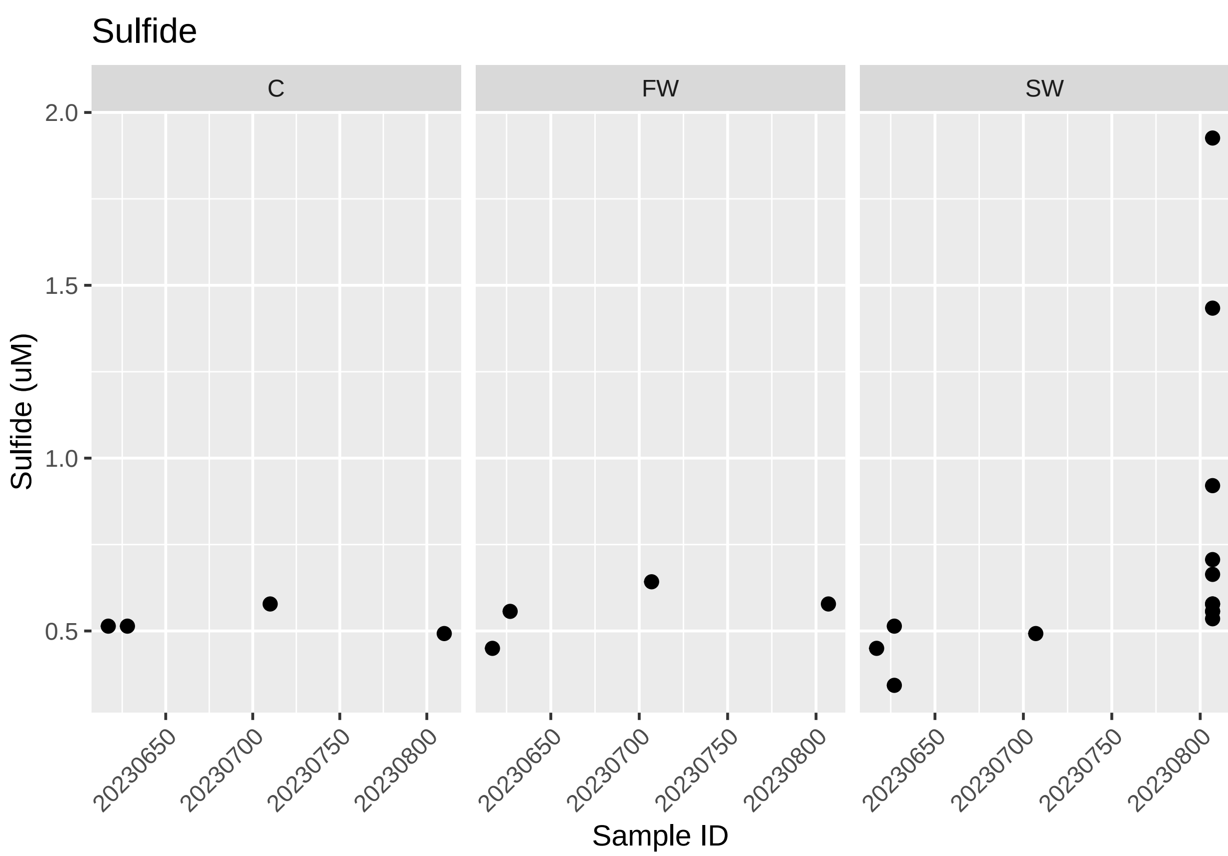
#CALCULATE CONCENTRATIONS

```r
samples<-left_join(plates,stds,by="Plate")%>% #add slope/intercept info
  filter(!str_detect(plates$IDs,"STD|DUP|SPK|ppt|spk|dup"))%>% #filter out standards and checks
  mutate(conc_uM=((ABS1-intercept)/slope)*Dilutions)%>% #calculate concentrations
  mutate(conc_uM=ifelse(conc_uM<0,0,conc_uM))%>% #make negative values 0
  group_by(IDs)%>%
  summarise(conc_uM=mean(conc_uM))%>%
  separate_wider_delim(IDs,names=c("CollectionDate","Treatment","Grid"),delim="_",cols_remove = FALSE)%
  mutate(CollectionDate=as.numeric(CollectionDate))
```

```r
head(samples)
```

```
## # A tibble: 6 x 5
##   CollectionDate Treatment Grid  IDs            conc_uM
##            <dbl> <chr>     <chr> <chr>            <dbl>
## 1       20230617 C         H3    20230617_C_H3    0.514
## 2       20230617 FW        C3    20230617_FW_C3   0.450
## 3       20230617 SW        B4    20230617_SW_B4   0.450
## 4       20230627 FW        D5    20230627_FW_D5   0.557
## 5       20230627 SW        D5    20230627_SW_D5   0.514
## 6       20230627 SW        F4    20230627_SW_F4   0.343
```

```r
p<-samples%>%
  ggplot(aes(x=CollectionDate,y=conc_uM))+
  geom_point(size=2)+
  facet_wrap(~Treatment,scales = "free_x")+
  labs(y="Sulfide (uM)",x="Sample ID")+
  theme(axis.text.x = element_text(angle=45,vjust=1,hjust=1),
        legend.position = "none")+
  ggtitle("Sulfide")

p
```



#EXPORT

```r
# setwd("C:/Users/ashmo/OneDrive - Arizona State University/Data/TEMPEST/Porewater/04_SummarizedData")

write.csv(samples, file="Processed Data/TEMPEST2023_H2S.csv")
```